

Woong Han

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Skills

Programming Language: C, C++, Python, Java, Go, JavaScript, HTML/CSS

Framework: CUDA, OpenMP, MPI, gem5, Yocto, OpenEmbedded, Pandas, NumPy, Scikit-learn

Tool: Docker, GDB, Valgrind, Wireshark, Oscilloscopes, Jira, Confluence, Git, Jenkins, Vim, Doxygen, VS Code, PlantUML, MobaXterms, Matplotlib, Seaborn, Jupyter Notebook, PuTTY **OS:** Linux, Windows

Experience

Software Engineer, LG Electronics – Seoul, South Korea

Apr 2022 – July 2025

- **Developed and implemented C/C++-based logging systems for 15+ modules in automotive** telematics devices and clusters to establish an **efficient and reliable debugging in TCP/IP environment**.
- **Contributed to an open-source project** (COVESA DLT-Daemon) by resolving a dead code issue related to mutex usage and implementing a new configuration feature.
- **Resolved 200+ technical cross-compiled Linux-embedded system issues** including module crashes, file system malfunctions, and performance deterioration (e.g. reduced CPU usage of a module by 5%).
- **Developed and implemented 10+ synchronous and asynchronous APIs** for applications in embedded systems, precisely meeting client requirements.
- Designed high-level software architecture using UML tools, enhancing system design clarity and documentation.
- **Collaborated with 10+ cross-functional LGE teams and global OEMs (Toyota and GM)**, to define requirements and conceptualize solutions, contributing to the successful deployment of complex systems.
- **Integrated OEM platforms and third-party software into the systems**, ensuring smooth progress in a CI/CD software development environment.

Education

University of Minnesota, MS in Computer Science

Sept 2025 – Present

- **GPA: 4.0/4.0**, Coursework: **Operating Systems, Intro. to Parallel Computing, Intro. to Distributed Systems**

Ajou University, BS in Electrical and Computer Engineering

Mar 2016 – Feb 2022

- **GPA: 4.25/4.5 (3rd out of 196)**, **Early Graduation (one semester early)**

Projects

Xv6 Operating Systems (RISC-V) Extension

Sept 2025 - Dec 2025

- Implemented multiple **kernel features** such as **Copy-On-Write fork, superpage, and symlink**, enhancing virtual memory efficiency, system scalability, and filesystem flexibility.
- **Extended E1000 NIC driver** to transmit and receive packets via DMA using descriptor ring buffers.
- **Debugged kernel panics** and memory violations using qemu-gdb.

N-Body Simulation Worklog

Sept 2025 - Dec 2025

- Accelerated N-Body simulations by implementing CUDA kernels leveraging **shared memory tiling, thread coarsening, hyperparameter tuning, hierarchical method, and CUTLASS** to maximize GPU utilization.

Parallel Algorithm Acceleration

Sept 2025 - Dec 2025

- **Parallelized Dynamic Time Warping using OpenMP** achieving significant speedups through multithreading on large datasets.
- Implemented a **sparse, block-cyclic variant of Cannon's distributed matrix multiplication in MPI**, reducing communication/computation overhead and improving load balancing.

Gem5 O3 CPU Performance Counter

Nov 2021 - Dec 2021

- Implemented a **performance counter in the gem5 Out-Of-Order CPU architecture** to apply the concept of Front Miss Table, improving skills in C++ and Python and deepening knowledge of CPU architecture.